ANNEXURE II

SUBMISSION OF PROJECT SYNOPSIS AND GUIDE ACCEPTANCE FORM

(To be submitted to the Project Steering Committee)

PART A: Project Synopsis

I. Student Details:

1. Name of the Program: Bachelor of Computer Applications (BCA)
2. Name of the Student: R RAMYASREE
3. Roll Number: 2414105526
4. Session & Year: JANUARY/FEBRUARY 2026

II. Project Synopsis:

1. **TITLE OF THE PROJECT:** WEB – BASED BILLING SYSTEM FOR SMALL RETAIL STORES.

2. **INTRODUCTION:**

   Small retail shops depend on manual billing methods, which results in calculation errors, delays and difficulty in maintaining sales records. As digital adoption increases, even small businesses require simple, affordable and user-friendly billing solutions. This project aims to design and develop a Web – Based Billing System that enables shop owners to generate bills quickly, store transaction details securely and manage product information efficiently.

   The system will simplify day – to – day operations by automating billing, applying taxes and generating sales summaries. It reduces dependency on manual registers and improves accuracy in financial records. The project focuses on creating an intuitive interface suitable for non-technical users, ensuring smooth adoption in small retail environments.

3. **OBJECTIVES OF THE STUDY:**
   - To design a simple and efficient online billing platform for small shops.
   - To automate bill generation with accurate tax calculations.
   - To maintain product, customer and sales data in a structured database.
   - To generate daily, weekly and monthly sales reports.
   - To reduce manual errors and improve operational efficiency.

4. LITERATURE REVIEW:

Digital billing systems have become essential tools for modern retail operations. Research shows that small businesses often face challenges with manual billing due to time consumption, human errors and lack of proper record – keeping. Studies on retail automation shows that computerized billing improves accuracy, enhances customer experience and supports better decision – making through data analytics. However, many available billing solutions are expensive, complex or designed for large enterprises, making them unsuitable for small shops with limited resources.

Research on small – scale retail operations indicates that shop owners prefer systems that are easy to use, require minimal training and offer essential features like product management, tax calculation and sales tracking. Cloud – based billing systems have gained popularity due to their accessibility and low maintenance requirements. Researchers also emphasize the importance of secure data storage and user – friendly interfaces in retail billing applications.

PROBLEM STATEMENT: Small retail shops commonly use manual billing methods, which lead to:

- Frequent calculation errors.
- Difficulty in tracking sales and inventory.
- Lack of digital records for audits or analysis.
- Time – consuming billing during peak hours.
- No centralized system for product and customer data.

There is a clear need for a simple, affordable and efficient online billing system tailored specifically for small retail shops. This project aims to bridge this gap by developing a lightweight, user – friendly solution that automates billing and improves operational efficiency.

Therefore, there is a need for a lightweight, cost – effective and easy – to – use web – based billing system tailored for small retail stores.

5. RESEARCH METHODOLOGY:

The project will follow a structured methodology to ensure systematic development and accurate results.

Requirement Analysis:

- Identify the needs of small retail shops through observation and informal discussions.
- Define core features: Product management, billing, tax calculation, sales reports and user authentication.
- Determine constraints such as low cost, ease of use and minimal hardware requirements.

System Design:

- Prepare Data Flow Diagrams to represent system processes.
- Design the database schema for products, customers, bills and sales records.
- Create wireframes for the intuitive user interface focusing on simplicity and clarity.

Technology Selection:

- Choose Java for backend logic due to its reliability and platform independence.
- Use MySQL for structured data storage.
- Implement a web –based interface using HTML, CSS and JavaScript for accessibility.

Development Phase:

- Implement modules: Login, Product Management, Billing, Tax calculation, Sales Reports.
- Integrate database operations using JDBC.
- Ensure proper validation and error handling.

Testing:

- Perform unit testing for each module.
- Conduct integration testing to ensure smooth interaction between components.
- Test usability with sample users to confirm ease of use.

Deployment and Documentation:

- Deploy the system on a local server or cloud environment.
- Prepare user manual and technical documentation.
- Collect feedback for future enhancements.

6. **TECHNOLOGY TO BE USED:** The online billing system will be developed using a combination of proven, widely used technologies suitable for small -scale applications.

Frontend technologies:

- HTML & CSS: For creating a clean and responsive user interface.
- JavaScript:For form validation, dynamic updates and improved user experience.

Backend Technology:

JAVA:

- Provides platform independence and strong security features.
- Supports modular programming, making the system easy to maintain.
- Ideal for building reliable billing logic and business rules.

Database:

MySQL:

- Efficient for handling structured data like products, bills and sales records.
- Supports SQL queries for fast data retrieval and reporting.
- Easy to integrate with Java using JDBC.

Connectivity:

JDBC (Java Database Connectivity):

- Enables communication between Java application and MySQL database.
- Ensures secure and efficient data transactions.

Deployment Environment:

- The system can run on any standard browser.
- Backend can be hosted on a local server or cloud platform.
- Minimal hardware requirements make it suitable for small shops.

Security Measures:

- Password – protected login system.
- Input validation to prevent incorrect entries.
- Secure database access with restricted privileges.

Technology Stack is needed because:

- Easy to implement and learn.
- Cost – effective and open source.
- Scalable for future enhancements like barcode scanning, inventory alerts or online payments.

7. **REFERENCES:**

- Oracle. (2024). Java Platform Standard Edition Documentation.
- Oracle. (2024). MySQL Reference Manual.
- Sommerville, I. (2016). Software Engineering (10th ed.).Pearson.
- Laudon, K.C., & Laudon, J.P.(2019). Management Information Systems.
- Online resources on small business digital transformation.
- Research articles on digital billing and retail automation from Google Scholar.
- Online resources and tutorials related to Java, JDBC and Web development.

III. Guide Details:

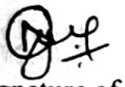
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4. Affiliation: ITRON INC.
5. Qualification: Bachelor of Technology
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PART – B: Guide Acceptance

1. I, Mr. Naveen Kumar Bellam working with ITRON INC. as Principal IT solutions delivery hereby confirm my willingness to guide Mrs. R Ramyasree Reg No. 2414105526 for the topic "Web – Based Billing System for Small Retail Stores." during the period June 2026 to October 2026.

Place: 7501 Pearson Ranch Road, Unit 7201 Austin, Texas – 78717 US

Date: 27th June, 2026



(Signature of the Guide)

(Note: A Guide needs to get registered with the University if he/ she is guiding a MUJDOE project for the first time. The guide Registration form can be downloaded from the LMS portal)

DECLARATION

I hereby declare that this project synopsis is an original work carried out by me and has not been/will not be submitted to any other University for the fulfilment of any course of study.

Place: 7501 Pearson Ranch Road, Unit 7201 Austin, Texas – 78717 UNITED STATES.

Date: 27th June, 2026

(*Filled in application forms to be signed by both student and the Guide. Forms must be scanned in either .pdf/.doc format and submitted through the LMS student's Login.)



(Signature of the Student)